
Demo Contract

Robotic Intervention for Pollution and Plastic Litter Extraction (RIPPLE)

MECHTRON 4TB6: Mechatronics Capstone Design Process

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Group 14

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Project Description

The purpose of RIPPLE is to collect floating trash on the water surface of local ponds, specifically West Pond (located near McMaster University). The project is to design and manufacture an aquatic system (referred to as the *system*) that can maneuver on the water surface, collecting any floating trash. The system will collect the trash using a Trash Collection Mechanism (TCM). The system will be remotely controlled by the user and communicate its location back to the User Interface (UI). It will identify obstacles in its path and relay this information back to the user via a video feed.

Scope of Work (SOW)

The final version of RIPPLE will deliver the following functionalities:

1. Able to operate on the surface of the water while carrying the TCM and all associated electronics
 - a. Able to float on the surface of the water without tipping over while in motion
2. Able to collect floating trash via the TCM
3. Remotely controllable by the user
 - a. User will control the speed and direction of movement
 - b. Able to propel forwards
 - c. Able to turn right and left
4. Remotely locatable by user
 - a. Able to communicate its location (coordinates) to the UI
5. User will be able to identify obstacles and nearby trash in the system's path via a video feed